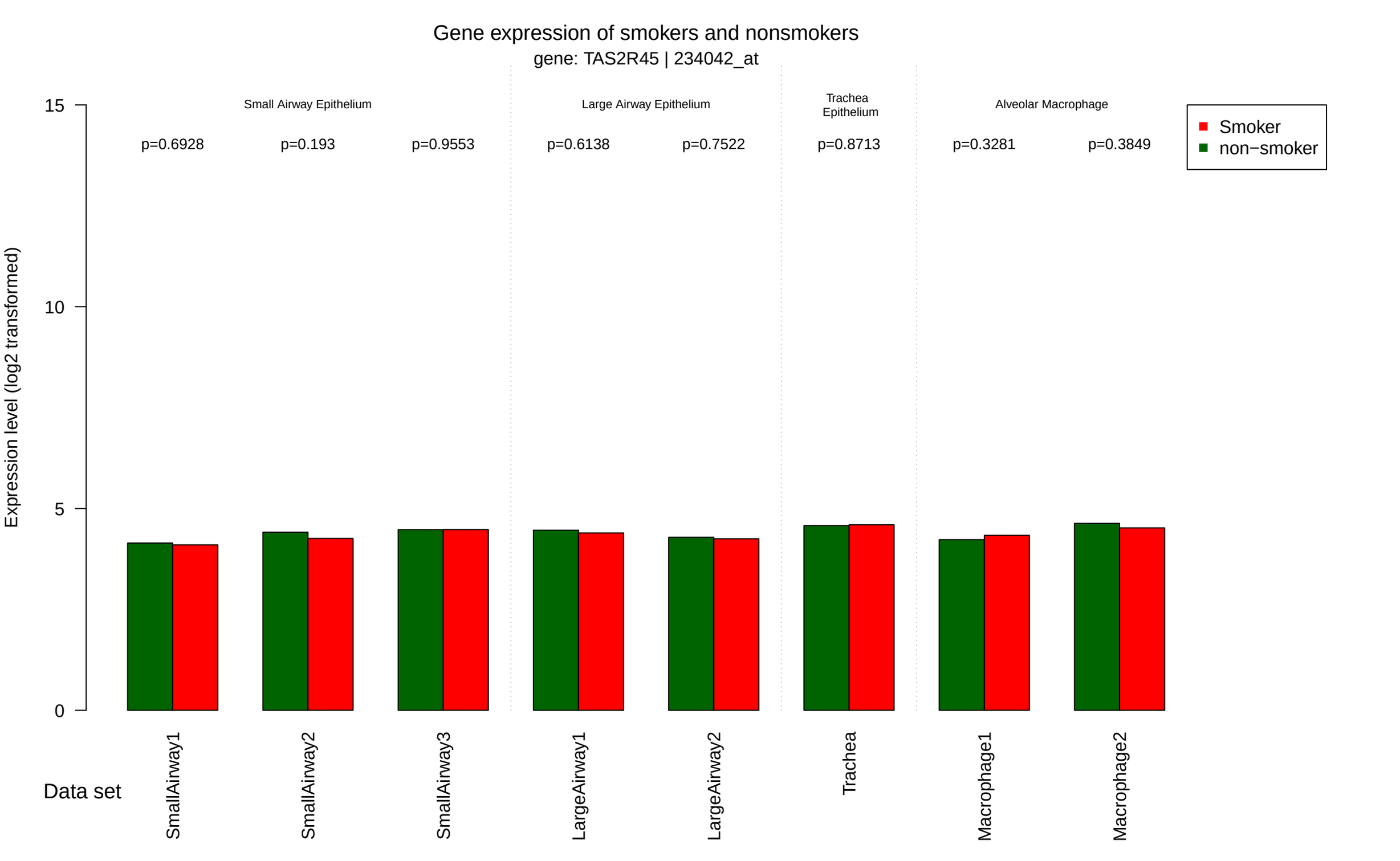
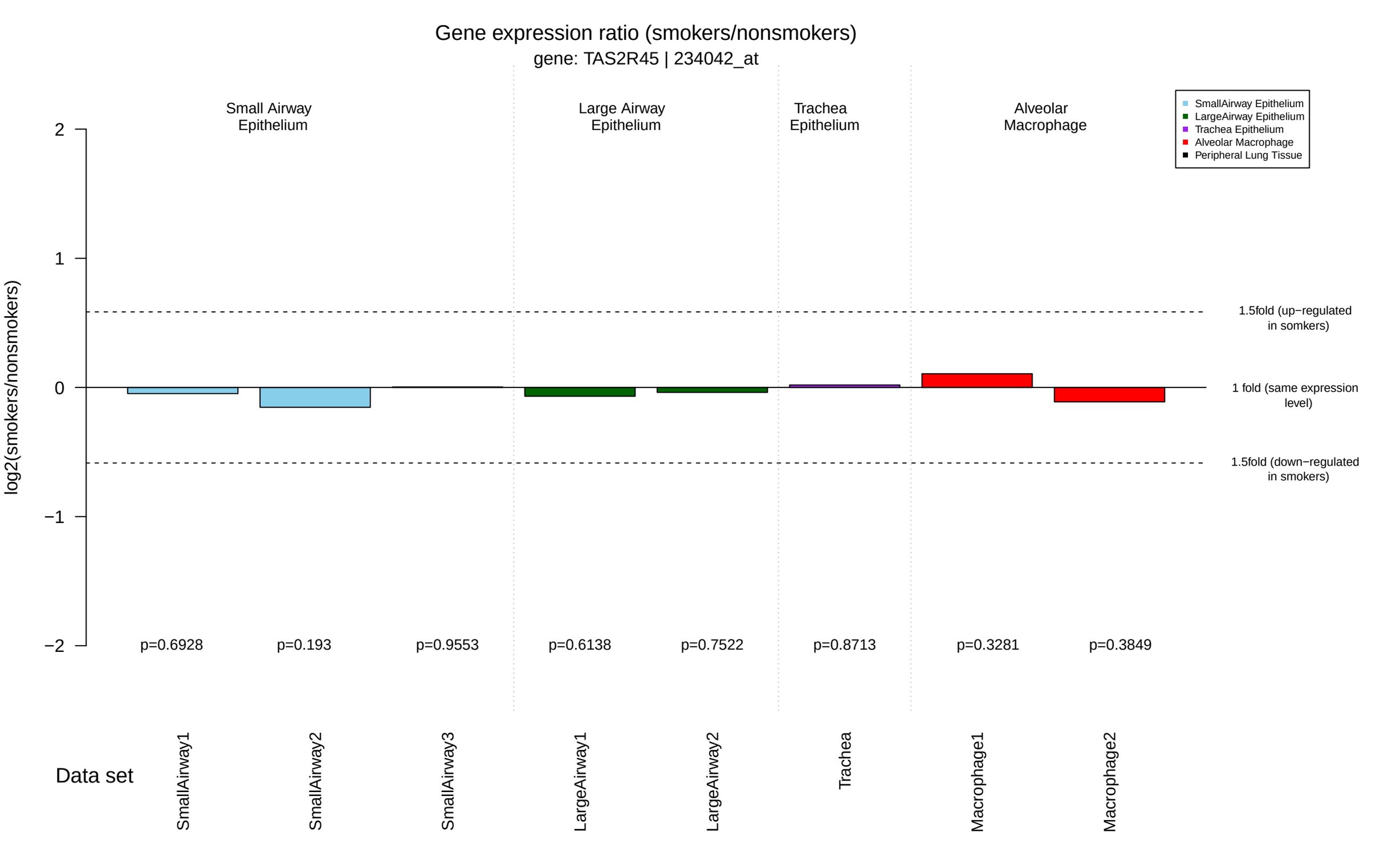
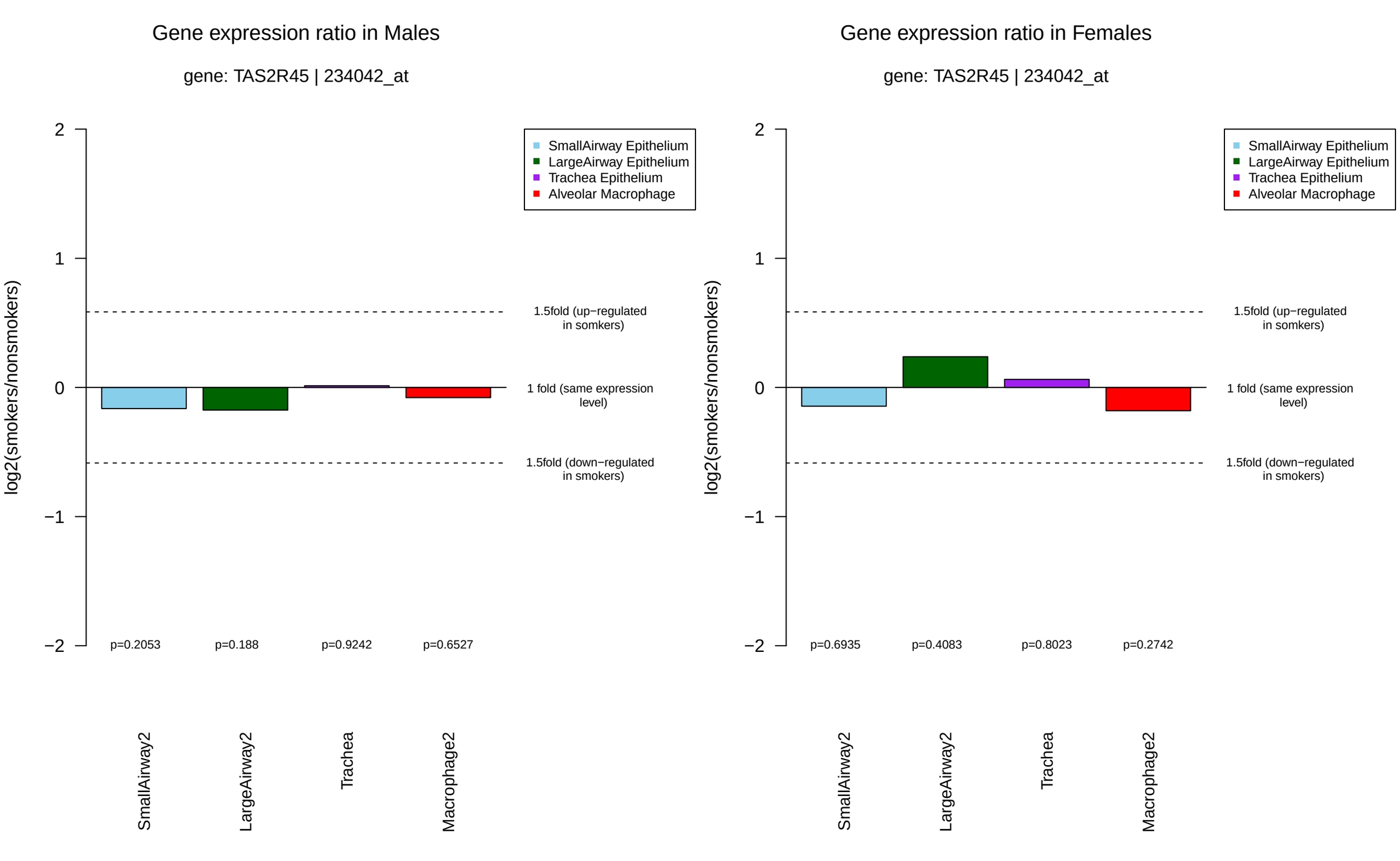
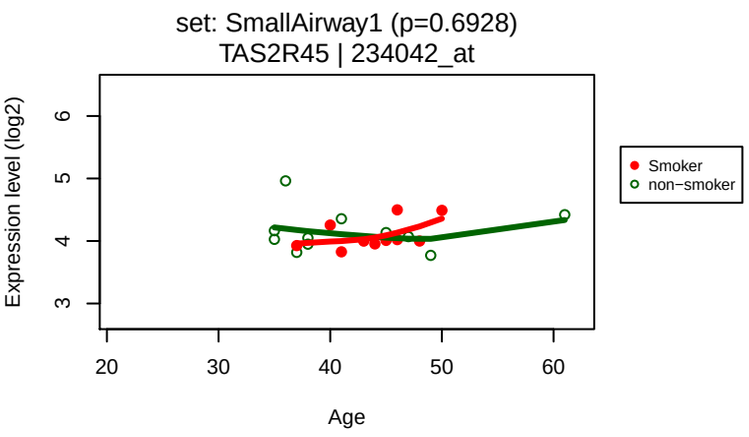
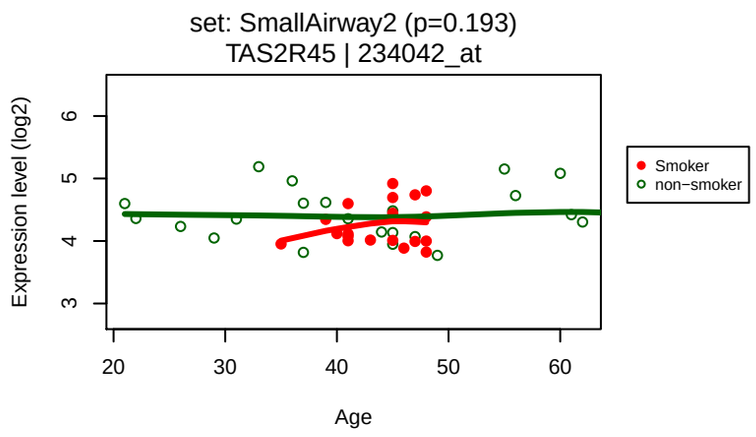
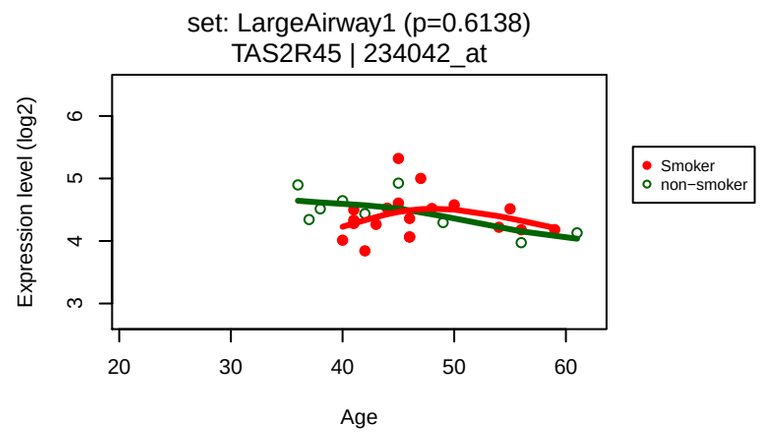

Gene expression of smokers and nonsmokers
gene: TAS2R45 | 234042_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6928
p=0.193
p=0.9553
p=0.6138
p=0.7522
p=0.8713
p=0.3281
p=0.3849
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

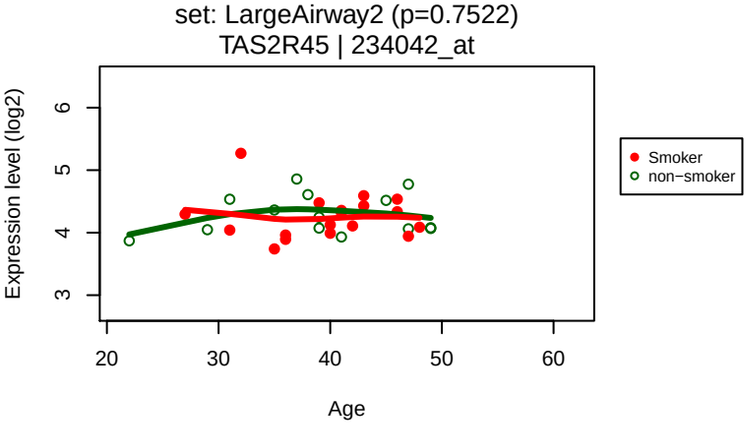

Gene expression ratio (smokers/nonsmokers)
gene: TAS2R45 | 234042_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6928
p=0.193
p=0.9553
p=0.6138
p=0.7522
p=0.8713
p=0.3281
p=0.3849
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

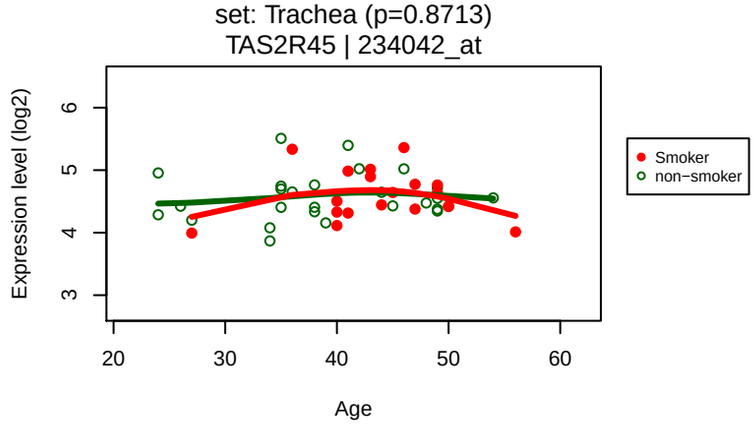

Gene expression ratio in Males
gene: TAS2R45 | 234042_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2053
p=0.188
p=0.9242
p=0.6527
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAS2R45 | 234042_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6935
p=0.4083
p=0.8023
p=0.2742
SmallAirway2
LargeAirway2
Trachea
Macrophage2

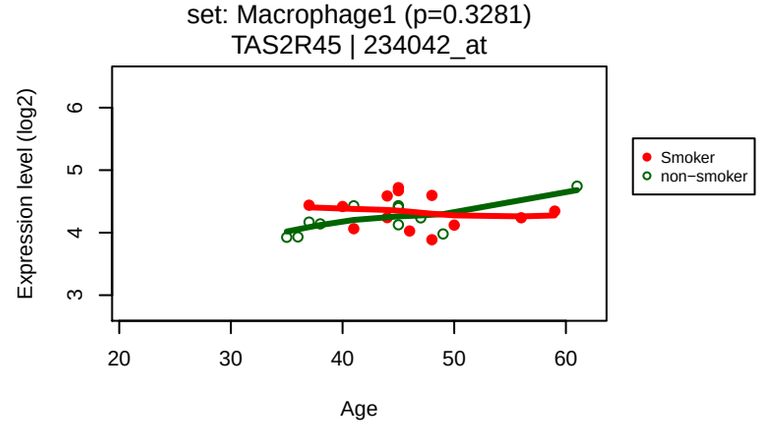

set: SmallAirway1 (p=0.6928)
TAS2R45 | 234042_at
Expression level (log2)
Age
Smoker
non−smoker

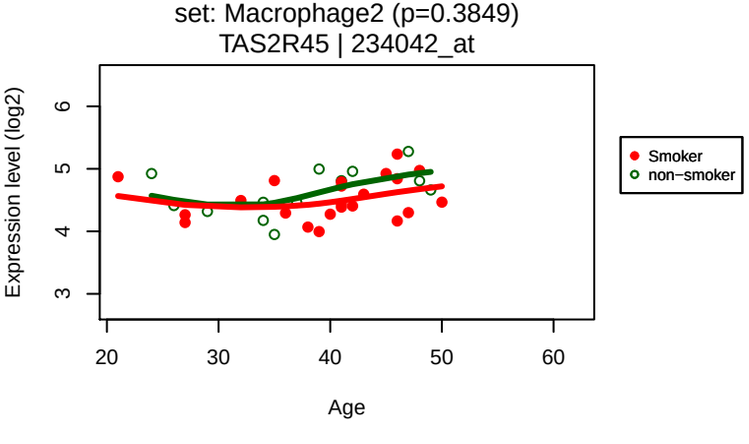

set: SmallAirway2 (p=0.193)
TAS2R45 | 234042_at
Expression level (log2)
Age
Smoker
non−smoker

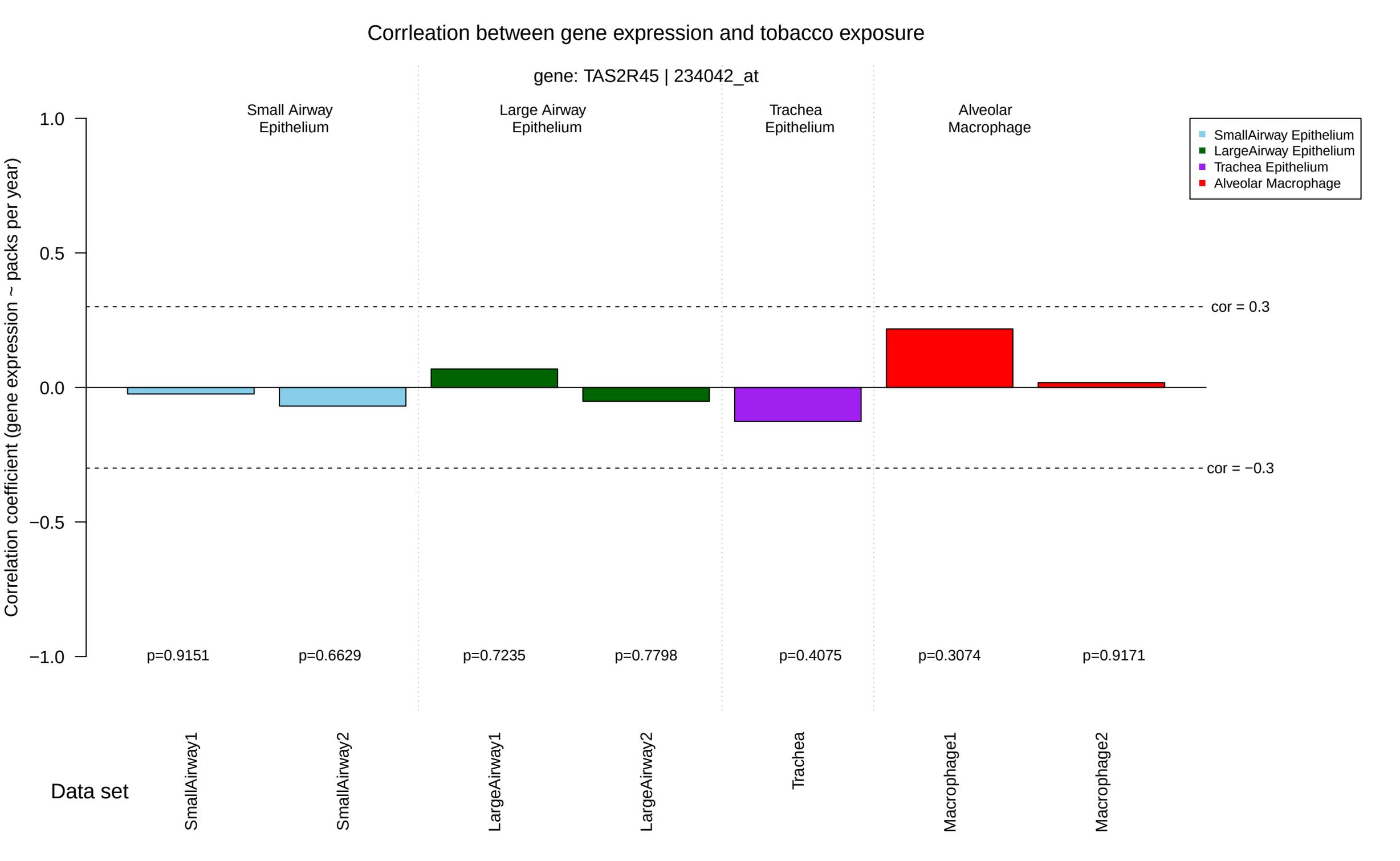

set: LargeAirway1 (p=0.6138)
TAS2R45 | 234042_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7522)
TAS2R45 | 234042_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8713)
TAS2R45 | 234042_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3281)
TAS2R45 | 234042_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3849)
TAS2R45 | 234042_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAS2R45 | 234042_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9151
p=0.6629
p=0.7235
p=0.7798
p=0.4075
p=0.3074
p=0.9171
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2